Mentl: a language whose compiler proves what software will never do

Hard Tech Week 2026 · Nucleus × Startup State open call · [[name]] · sonofagoose@pm.me · github.com/ampactor-labs/mentl

THE PROBLEM

AI now helps write the software that runs machines: batteries, flywheels, flight controls. People test that code and review it, and then they trust it. No mainstream language lets a team write "this never touches the internet" or "this never asks for more than 100 kW" and have the compiler prove it.

WHAT EXISTS

Mentl is a programming language I have built in Utah since March 15, 2026: 2,610 commits before this week, open source. You write a promise into the code, and the compiler checks it through every function the code calls. Break the promise and the program won't build. The compiler is written in Mentl and compiles itself.

sonido is my audio library, built since January 25, 2026: effects that run in plugins, in a browser and on a guitar pedal.

BUILT THIS WEEK

- A guitar pickup hears the power grid's frequency in a room's wiring, measured by sonido to a thousandth of a hertz and shown on sonido's tuner.
- A model of a Torus Station decides, typically in under a millisecond, what its flywheel and battery should do: ride out tomorrow's rush hour, catch a real 2019 grid failure, keep the building on when the grid goes dark.
- That decision is written in Mentl and proven unable to reach the internet. Add one line that reads a price online and the compiler refuses to build it.

WHO NEEDS IT IN UTAH

- **Torus**, Salt Lake City: its Stations steady the grid, and the software inside each one is where a proof would matter first. Torus's judge is seeing this demo today. First conversation: Torus engineering, this month.
- **Hill AFB, 309th Software Engineering Group**: maintains flight software for the F-16, the F-35 and the missile fleet, where every claim about the software needs evidence.

NINETY DAYS

Day 30	Form the Utah company and register for federal grants, which takes weeks. Run the controller on a battery inverter on my bench.
Day 60	Run it beside a real battery at one Utah site, alongside the site's own controls.
Day 90	A pilot agreement with that site, and an SBIR proposal drafted with Nucleus Grow.

THE $10,000

$7,000	a battery inverter and battery for the bench, so the controller drives real power hardware
$3,000	forming the company, federal grant registration, and installing at the pilot site

WHAT I NEED

- A Utah battery or flywheel operator who will host the pilot.
- An introduction to the 309th, through 47G, and Nucleus Grow's help with the SBIR proposal.
- A co-founder who knows power electronics.

The first business is paid pilots with operators of batteries and flywheels. The second is Mentl itself, as the language that teams and their AI assistants write such code in. The demo: hum.ampactor.dev, and its source at github.com/ampactor-labs/hum. [[one line: who you are, in your words]]